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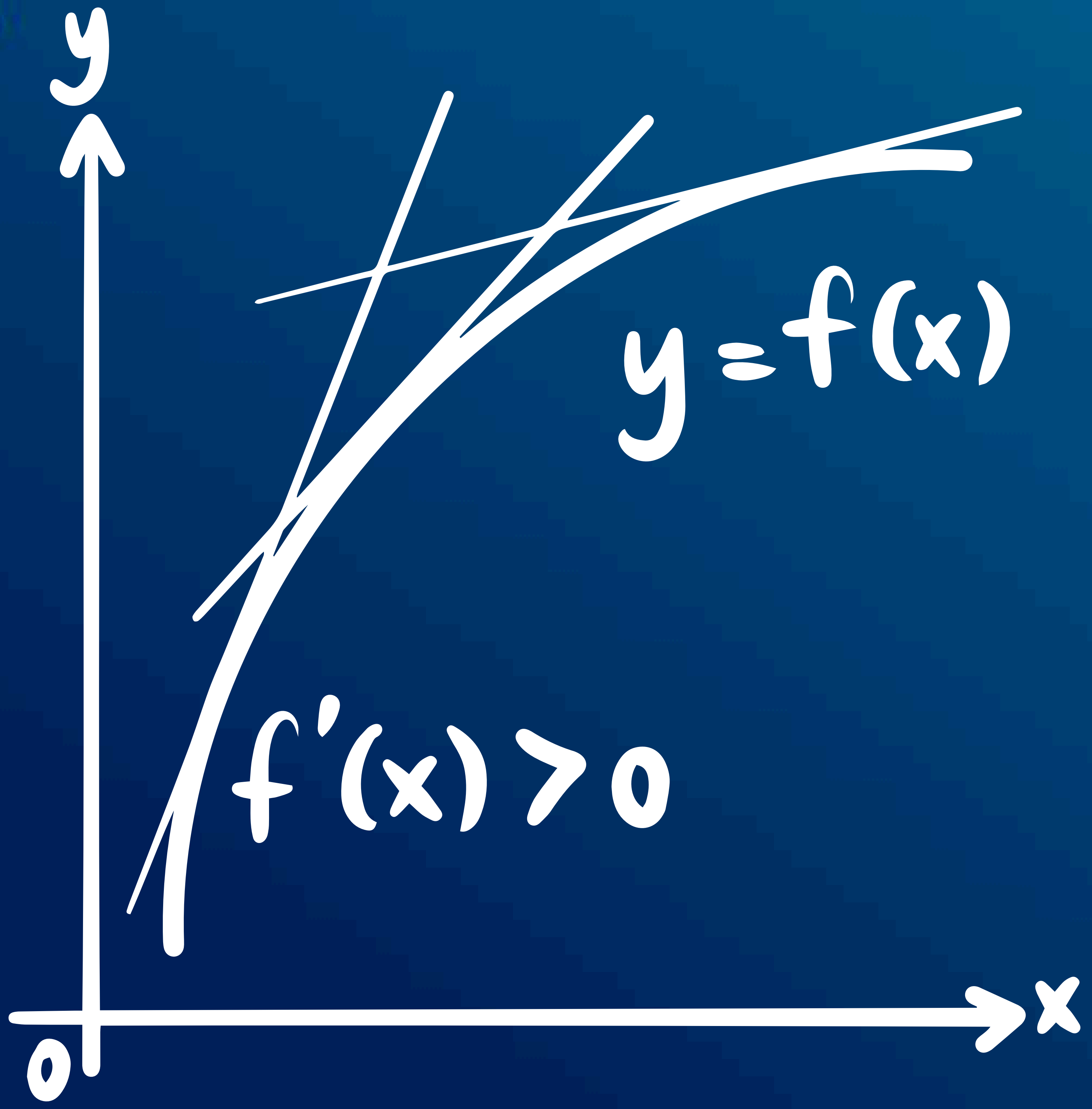

Calculus 1

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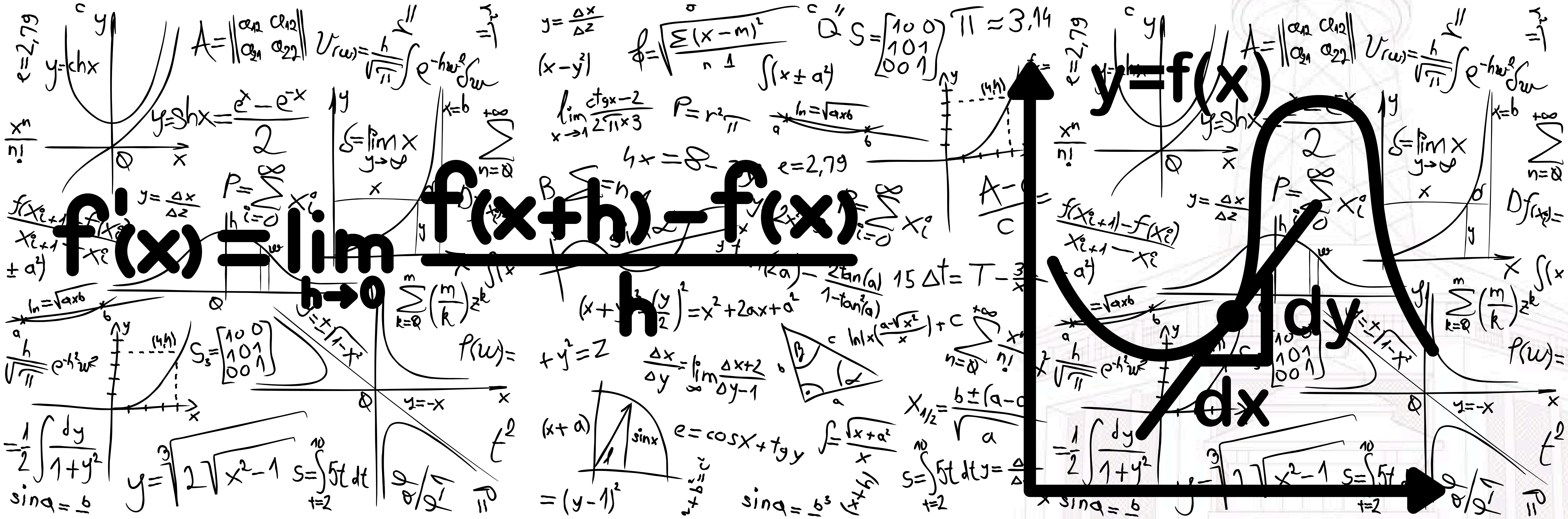
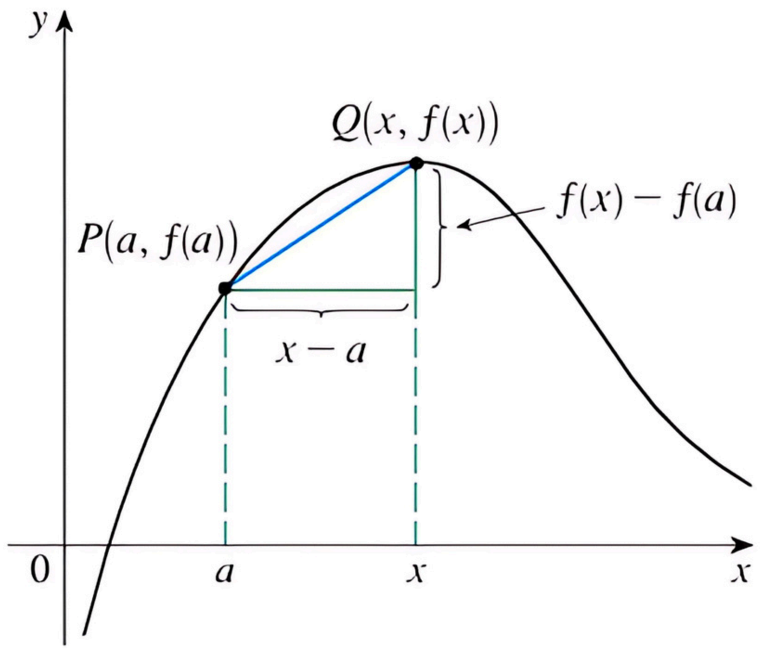
THE DERIVATIVE :

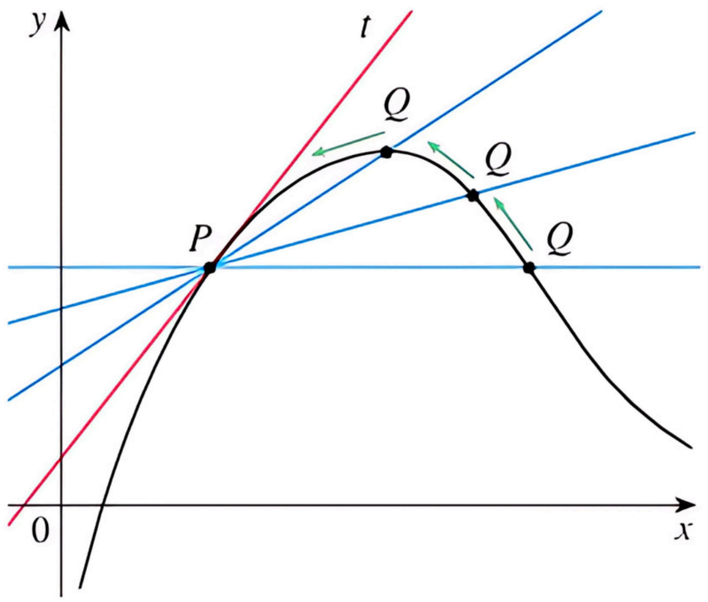
Definition

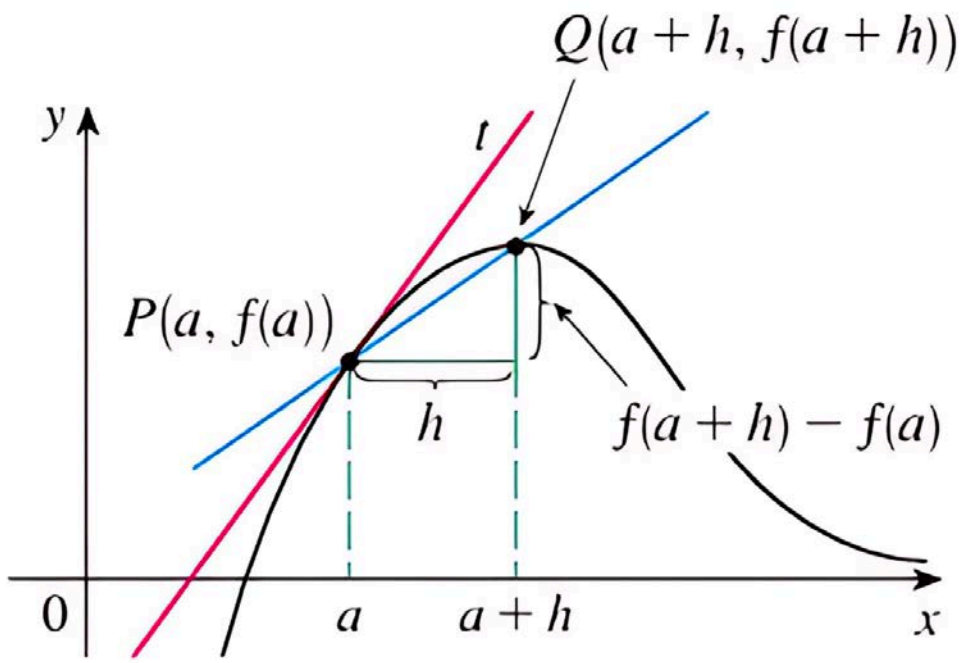


Derivative

Tangent Line







Tangents

$$m_{PQ} = \frac{f(x) - f(a)}{x - a}$$

1 Definition The **tangent line** to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope

$$m = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

provided that this limit exists.

$$m = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}$$

Derivative

Definition & Notation

Definition Derivative

The **derivative** of a function f is another function f' (read "f prime") whose value at any number x is

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \qquad f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

If this limit does exist, we say that f is **differentiable** at x. Finding a derivative is called **differentiation**; the part of calculus associated with the derivative is called **differential calculus.**

Notations

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x) = Df(x) = D_x f(x)$$

Derivative

Definition

EXAMPLE 1 Let $f(x) = 13x - 6$. Find $f'(4)$.

SOLUTION

$$f'(4) = \lim_{h\to 0} \frac{f(4+h) - f(4)}{h} = \lim_{h\to 0} \frac{[13(4+h) - 6] - [13(4) - 6]}{h}$$

$$= \lim_{h\to 0} \frac{13h}{h} = \lim_{h\to 0} 13 = 13$$

EXAMPLE 2 If $f(x) = x^3 + 7x$, find $f'(x)$.

SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h\to 0} \frac{\left[(x+h)^3 + 7(x+h)\right] - \left[x^3 + 7x\right]}{h} \\ &= \lim_{h\to 0} \frac{3x^2h + 3xh^2 + h^3 + 7h}{h} \\ &= \lim_{h\to 0} (3x^2 + 3xh + h^2 + 7) \\ &= 3x^2 + 7 \end{aligned}$$

EXAMPLE 3 If $f(x) = 1/x$, find $f'(x)$.

SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{\dfrac{1}{x+h} - \dfrac{1}{x}}{h} \\ &= \lim_{h\to 0} \left[\frac{x - (x+h)}{(x+h)x} \cdot \frac{1}{h}\right] = \lim_{h\to 0} \left[\frac{-h}{(x+h)x} \cdot \frac{1}{h}\right] \\ &= \lim_{h\to 0} \frac{-1}{(x+h)x} = -\frac{1}{x^2} \end{aligned}$$

Thus, f' is the function given by $f'(x) = -1/x^2$. Its domain is all real numbers except $x = 0$. ■

Derivative

Definition

EXAMPLE 4 Find $F'(x)$ if $F(x) = \sqrt{x}, x > 0$.

SOLUTION

$$F'(x) = \lim_{h\to 0} \frac{F(x+h) - F(x)}{h}$$

$$= \lim_{h\to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

$$F'(x) = \lim_{h\to 0} \left[\frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}\right]$$

$$= \lim_{h\to 0} \frac{x+h-x}{h\left(\sqrt{x+h} + \sqrt{x}\right)}$$

$$= \lim_{h\to 0} \frac{h}{h\left(\sqrt{x+h} + \sqrt{x}\right)}$$

$$= \lim_{h\to 0} \frac{1}{\sqrt{x+h} + \sqrt{x}}$$

$$= \frac{1}{\sqrt{x} + \sqrt{x}} = \frac{1}{2\sqrt{x}}$$

Thus, F', the derivative of F, is given by $F'(x) = 1/\left(2\sqrt{x}\right)$. Its domain is $(0, \infty)$. ■

EXAMPLE 5 Use the last boxed result to find $g'(c)$ if $g(x) = 2/(x+3)$.

SOLUTION

$$g'(c) = \lim_{x\to c} \frac{g(x) - g(c)}{x - c} = \lim_{x\to c} \frac{\dfrac{2}{x+3} - \dfrac{2}{c+3}}{x - c}$$

$$= \lim_{x\to c} \left[\frac{2(c+3) - 2(x+3)}{(x+3)(c+3)} \cdot \frac{1}{x-c}\right]$$

$$= \lim_{x\to c} \left[\frac{-2(x-c)}{(x+3)(c+3)} \cdot \frac{1}{x-c}\right]$$

$$= \lim_{x\to c} \frac{-2}{(x+3)(c+3)} = \frac{-2}{(c+3)^2}$$

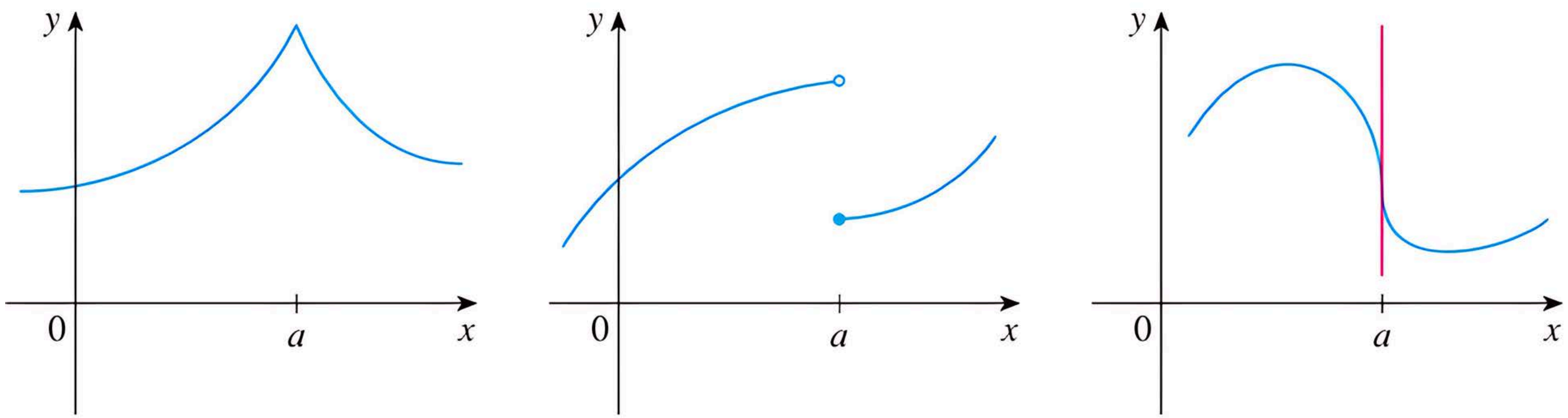
Derivative

Differentiable

Theorem A **Differentiability Implies Continuity**

If $f'(c)$ exists, then f is continuous at c.

Fail To Be Differentiable?



(a) A corner (b) A discontinuity (c) A vertical tangent

Derivative

Differentiable

EXAMPLE 6

Show that the function $f(x) = |x - 6|$ is not differentiable at 6. Find a formula for f' and sketch its graph.

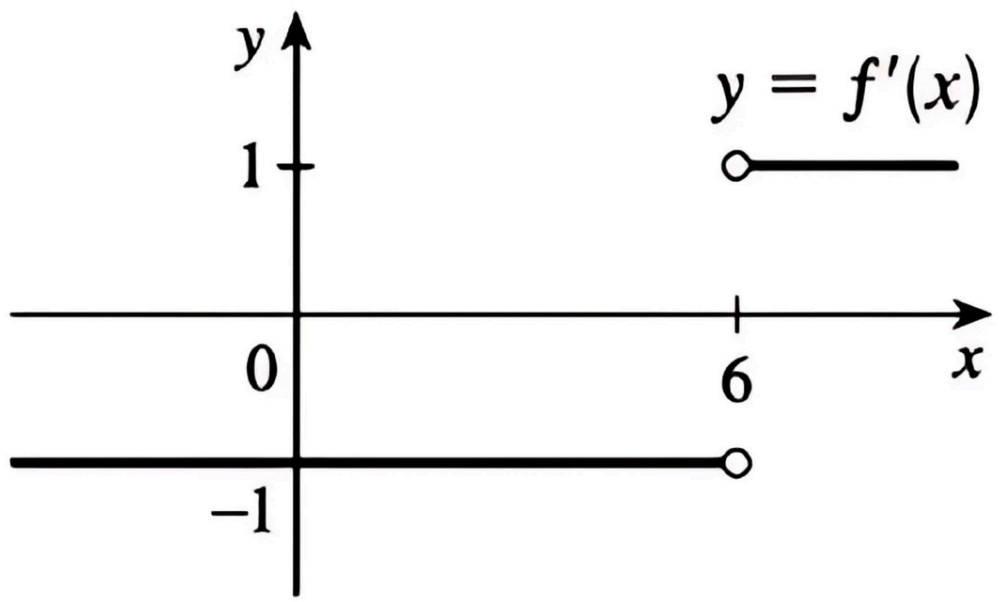
SOLUTION

$$f(x) = |x-6| = \begin{cases} x-6 & \text{if } x-6 \geq 0 \\ -(x-6) & \text{if } x-6 < 0 \end{cases} = \begin{cases} x-6 & \text{if } x \geq 6 \\ 6-x & \text{if } x < 6 \end{cases}$$

So the right-hand limit is $\displaystyle\lim_{x\to 6+} \frac{f(x)-f(6)}{x-6} = \lim_{x\to 6+} \frac{|x-6|-0}{x-6} = \lim_{x\to 6+} \frac{x-6}{x-6} = \lim_{x\to 6+} 1 = 1$, and the left-hand limit is $\displaystyle\lim_{x\to 6-} \frac{f(x)-f(6)}{x-6} = \lim_{x\to 6-} \frac{|x-6|-0}{x-6} = \lim_{x\to 6-} \frac{6-x}{x-6} = \lim_{x\to 6-} (-1) = -1$. Since these limits are not equal,

$\displaystyle f'(6) = \lim_{x\to 6} \frac{f(x)-f(6)}{x-6}$ does not exist and f is not differentiable at 6.

However, a formula for f' is $f'(x) = \begin{cases} 1 & \text{if } x > 6 \\ -1 & \text{if } x < 6 \end{cases}$



Derivative

Theorem

Theorem A Constant Function Rule

If $f(x) = k$, where k is a constant, then for any x, $f'(x) = 0$; that is,

$$D_x(k) = 0$$

Theorem B Identity Function Rule

If $f(x) = x$, then $f'(x) = 1$; that is,

$$D_x(x) = 1$$

Theorem C Power Rule

If $f(x) = x^n$, where n is a positive integer, then $f'(x) = nx^{n-1}$; that is,

$$D_x(x^n) = nx^{n-1}$$

Theorem D Constant Multiple Rule

If k is a constant and f is a differentiable function, then $(kf)'(x) = k \cdot f'(x)$; that is,

$$D_x[k \cdot f(x)] = k \cdot D_x f(x)$$

In words, *a constant multiplier k can be passed across the operator D_x.*

Theorem E Sum Rule

If f and g are differentiable functions, then $(f + g)'(x) = f'(x) + g'(x)$; that is,

$$D_x[f(x) + g(x)] = D_x f(x) + D_x g(x)$$

In words, *the derivative of a sum is the sum of the derivatives.*

Theorem F Difference Rule

If f and g are differentiable functions, then $(f - g)'(x) = f'(x) - g'(x)$; that is,

$$D_x[f(x) - g(x)] = D_x f(x) - D_x g(x)$$

Derivative

Theorem

Theorem G **Product Rule**

If f and g are differentiable functions, then

$$(f \cdot g)'(x) = f(x)g'(x) + g(x)f'(x)$$

That is,

$$D_x\left[f(x)g(x)\right] = f(x)D_xg(x) + g(x)D_xf(x)$$

Theorem H **Quotient Rule**

Let f and g be differentiable functions with $g(x) \neq 0$. Then

$$\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{g^2(x)}$$

That is,

$$D_x\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)D_xf(x) - f(x)D_xg(x)}{g^2(x)}$$

Derivative

Theorem

EXAMPLE 1 Find the derivatives of $5x^2 + 7x - 6$ and $4x^6 - 3x^5 - 10x^2 + 5x + 16$.

SOLUTION

$$\begin{aligned}
D_x(5x^2 + 7x - 6) &= D_x(5x^2 + 7x) - D_x(6) && \text{(Theorem F)}\\
&= D_x(5x^2) + D_x(7x) - D_x(6) && \text{(Theorem E)}\\
&= 5D_x(x^2) + 7D_x(x) - D_x(6) && \text{(Theorem D)}\\
&= 5 \cdot 2x + 7 \cdot 1 - 0 && \text{(Theorems C, B, A)}\\
&= 10x + 7
\end{aligned}$$

To find the next derivative, we note that the theorems on sums and differences extend to any finite number of terms. Thus,

$$\begin{aligned}
&D_x(4x^6 - 3x^5 - 10x^2 + 5x + 16)\\
&\quad = D_x(4x^6) - D_x(3x^5) - D_x(10x^2) + D_x(5x) + D_x(16)\\
&\quad = 4D_x(x^6) - 3D_x(x^5) - 10D_x(x^2) + 5D_x(x) + D_x(16)\\
&\quad = 4(6x^5) - 3(5x^4) - 10(2x) + 5(1) + 0\\
&\quad = 24x^5 - 15x^4 - 20x + 5
\end{aligned}$$

Derivative

Theorem

EXAMPLE 2 Let $g(x) = x, h(x) = 1 + 2x,$ and $f(x) = g(x) \cdot h(x) = x(1 + 2x)$. Find $D_x f(x), D_x g(x)$, and $D_x h(x)$, and show that $D_x f(x) \neq [D_x g(x)][D_x h(x)]$.

SOLUTION

$$\begin{aligned} D_x f(x) &= D_x[x(1 + 2x)] \\ &= D_x(x + 2x^2) \\ &= 1 + 4x \\ D_x g(x) &= D_x x = 1 \\ D_x h(x) &= D_x(1 + 2x) = 2 \end{aligned}$$

Notice that

$$D_x(g(x))D_x(h(x)) = 1 \cdot 2 = 2$$

whereas

$$D_x f(x) = D_x[g(x)h(x)] = 1 + 4x$$

Thus, $D_x f(x) \neq [D_x g(x)][D_x h(x)]$.

Derivative

Theorem

EXAMPLE 3 Find the derivative of $(3x^2 - 5)(2x^4 - x)$ by use of the Product Rule. Check the answer by doing the problem a different way.

SOLUTION

$$\begin{aligned} D_x\left[(3x^2 - 5)(2x^4 - x)\right] &= (3x^2 - 5)D_x(2x^4 - x) + (2x^4 - x)D_x(3x^2 - 5) \\ &= (3x^2 - 5)(8x^3 - 1) + (2x^4 - x)(6x) \\ &= 24x^5 - 3x^2 - 40x^3 + 5 + 12x^5 - 6x^2 \\ &= 36x^5 - 40x^3 - 9x^2 + 5 \end{aligned}$$

To check, we first multiply and then take the derivative.

$$(3x^2 - 5)(2x^4 - x) = 6x^6 - 10x^4 - 3x^3 + 5x$$

$$\begin{aligned} D_x\left[(3x^2 - 5)(2x^4 - x)\right] &= D_x(6x^6) - D_x(10x^4) - D_x(3x^3) + D_x(5x) \\ &= 36x^5 - 40x^3 - 9x^2 + 5 \end{aligned}$$

Derivative

Theorem

EXAMPLE 4 Find $\dfrac{d}{dx}\dfrac{(3x-5)}{(x^2+7)}$.

SOLUTION

$$\begin{aligned}\frac{d}{dx}\left[\frac{3x-5}{x^2+7}\right] &= \frac{(x^2+7)\dfrac{d}{dx}(3x-5) - (3x-5)\dfrac{d}{dx}(x^2+7)}{(x^2+7)^2}\\ &= \frac{(x^2+7)(3)-(3x-5)(2x)}{(x^2+7)^2}\\ &= \frac{-3x^2+10x+21}{(x^2+7)^2}\end{aligned}$$

EXAMPLE 5 Find $D_x y$ if $y = \dfrac{2}{x^4+1} + \dfrac{3}{x}$.

SOLUTION

$$\begin{aligned}D_x y &= D_x\left(\frac{2}{x^4+1}\right) + D_x\left(\frac{3}{x}\right)\\ &= \frac{(x^4+1)D_x(2) - 2D_x(x^4+1)}{(x^4+1)^2} + \frac{xD_x(3) - 3D_x(x)}{x^2}\\ &= \frac{(x^4+1)(0)-(2)(4x^3)}{(x^4+1)^2} + \frac{(x)(0)-(3)(1)}{x^2}\\ &= \frac{-8x^3}{(x^4+1)^2} - \frac{3}{x^2}\end{aligned}$$

Derivative

Trigonometric

Theorem A

The functions $f(x) = \sin x$ and $g(x) = \cos x$ are both differentiable and,

$$D_x(\sin x) = \cos x \qquad D_x(\cos x) = -\sin x$$

EXAMPLE 1 Find $D_x(3 \sin x - 2 \cos x)$.

SOLUTION

$$\begin{aligned} D_x(3 \sin x - 2 \cos x) &= 3D_x(\sin x) - 2D_x(\cos x) \\ &= 3 \cos x + 2 \sin x \end{aligned}$$

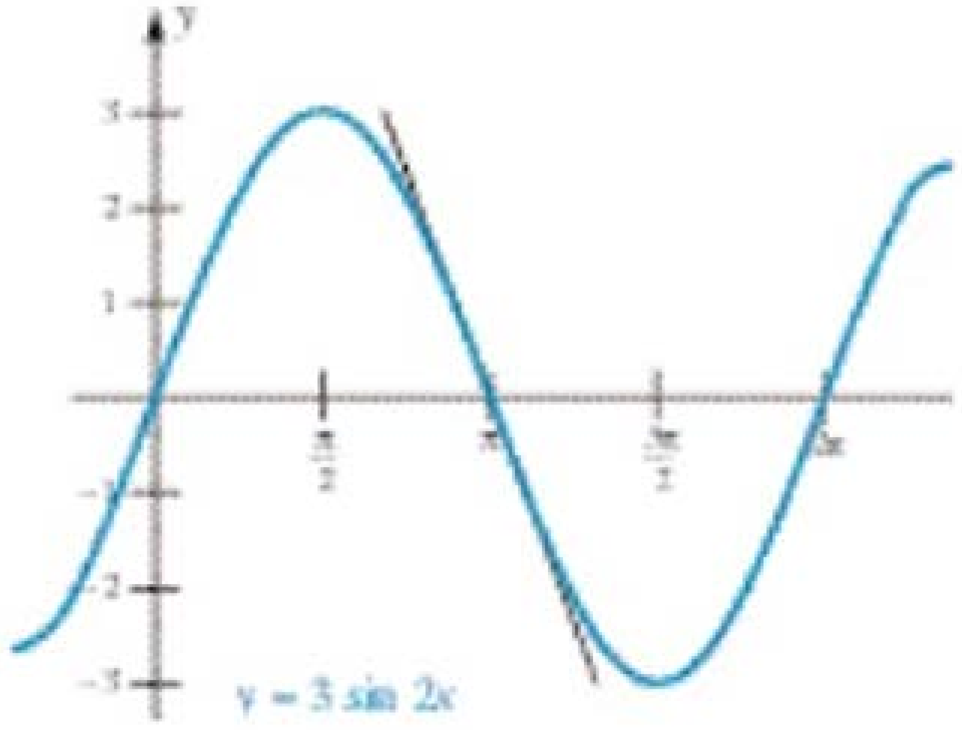


Figure 2

EXAMPLE 2 Find the equation of the tangent line to the graph of $y = 3 \sin x$ at the point $(\pi, 0)$. (See Figure 2.)

SOLUTION The derivative is $\frac{dy}{dx} = 3 \cos x$, so when $x = \pi$, the slope is $3 \cos \pi = -3$. Using the point-slope form for a line we find that the equation of the tangent line is

$$\begin{aligned} y - 0 &= -3(x - \pi) \\ y &= -3x + 3\pi \end{aligned}$$

Derivative

Trigonometric

EXAMPLE 3 Find $D_x(x^2 \sin x)$.

SOLUTION The Product Rule is needed here.

$$D_x(x^2 \sin x) = x^2 D_x(\sin x) + \sin x(D_x x^2) = x^2 \cos x + 2x \sin x$$

EXAMPLE 4 Find $\frac{d}{dx}\left(\frac{1 + \sin x}{\cos x}\right)$.

SOLUTION For this problem, the Quotient Rule is needed.

$$\frac{d}{dx}\left(\frac{1 + \sin x}{\cos x}\right) = \frac{\cos x\left(\frac{d}{dx}(1 + \sin x)\right) - (1 + \sin x)\left(\frac{d}{dx}\cos x\right)}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin x + \sin^2 x}{\cos^2 x}$$

$$= \frac{1 + \sin x}{\cos^2 x}$$

EXAMPLE 5 At time t seconds, the center of a bobbing cork is $y = 2 \sin t$ centimeters above (or below) water level. What is the velocity of the cork at $t = 0, \pi/2, \pi$?

SOLUTION The velocity is the derivative of position, and $\frac{dy}{dt} = 2 \cos t$. Thus, when $t = 0$, $\frac{dy}{dt} = 2 \cos 0 = 2$, when $t = \pi/2$, $\frac{dy}{dt} = 2 \cos \frac{\pi}{2} = 0$, and when $t = \pi$, $\frac{dy}{dt} = 2 \cos \pi = -2$. ■

Derivative

Trigonometric

Theorem B

For all points x in the function's domain,

$$D_x \tan x = \sec^2 x \qquad D_x \cot x = -\csc^2 x$$
$$D_x \sec x = \sec x \tan x \qquad D_x \csc x = -\csc x \cot x$$

EXAMPLE 6 Find $D_x(x^n \tan x)$ for $n \geq 1$.

SOLUTION We apply the Product Rule along with Theorem B.

$$\begin{aligned} D_x(x^n \tan x) &= x^n D_x(\tan x) + \tan x(D_x x^n) \\ &= x^n \sec^2 x + nx^{n-1} \tan x \end{aligned}$$

■

EXAMPLE 7 Find the equation of the tangent line to the graph of $y = \tan x$ the point $(\pi/4, 1)$.

SOLUTION The derivative of $y = \tan x$ is $\frac{dy}{dx} = \sec^2 x$. When $x = \pi/4$, the derivative is equal to $\sec^2 \frac{\pi}{4} = \left(\frac{2}{\sqrt{2}}\right)^2 = 2$. Thus the required line has slope 2 and passes through $(\pi/4, 1)$. Thus

$$y - 1 = 2\left(x - \frac{\pi}{4}\right)$$

$$y = 2x - \frac{\pi}{2} + 1$$

■

Derivative

Chain Rule

Theorem A Chain Rule

Let $y = f(u)$ and $u = g(x)$. If g is differentiable at x and f is differentiable at $u = g(x)$, then the composite function $f \circ g$, defined by $(f \circ g)(x) = f(g(x))$, is differentiable at x and

$$(f \circ g)'(x) = f'(g(x))g'(x)$$

That is,

$$D_x(f(g(x)) = f'(g(x))g'(x)$$

or

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

Derivative

Chain Rule

EXAMPLE 1 If $y = (2x^2 - 4x + 1)^{60}$, find $D_x y$.

SOLUTION We think of y as the 60th power of a function of x; that is

$$y = u^{60} \quad \text{and} \quad u = 2x^2 - 4x + 1$$

The outer function is $f(u) = u^{60}$ and the inner function is $u = g(x) = 2x^2 - 4x + 1$. Thus,

$$\begin{aligned} D_x y &= D_x f(g(x)) \\ &= f'(u)g'(x) \\ &= (60u^{59})(4x - 4) \\ &= 60(2x^2 - 4x + 1)^{59}(4x - 4) \end{aligned}$$

■

EXAMPLE 2 If $y = 1/(2x^5 - 7)^3$, find $\frac{dy}{dx}$.

SOLUTION Think of it this way.

$$y = \frac{1}{u^3} = u^{-3} \quad \text{and} \quad u = 2x^5 - 7$$

Thus,

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du}\frac{du}{dx} \\ &= (-3u^{-4})(10x^4) \\ &= \frac{-3}{u^4} \cdot 10x^4 \\ &= \frac{-30x^4}{(2x^5 - 7)^4} \end{aligned}$$

Derivative

Chain Rule

EXAMPLE 3 Find $D_t\left(\dfrac{t^3 - 2t + 1}{t^4 + 3}\right)^{13}$.

SOLUTION The last step in calculating this expression would be to raise the expression on the inside to the power 13. Thus, we begin by applying the Chain Rule to the function $y = u^{13}$, where $u = (t^3 - 2t + 1)/(t^4 + 3)$. The Chain Rule followed by the Quotient Rule gives

$$\begin{aligned}
D_t\left(\frac{t^3 - 2t + 1}{t^4 + 3}\right)^{13} &= 13\left(\frac{t^3 - 2t + 1}{t^4 + 3}\right)^{13-1} D_t\left(\frac{t^3 - 2t + 1}{t^4 + 3}\right) \\
&= 13\left(\frac{t^3 - 2t + 1}{t^4 + 3}\right)^{12} \frac{(t^4 + 3)(3t^2 - 2) - (t^3 - 2t + 1)(4t^3)}{(t^4 + 3)^2} \\
&= 13\left(\frac{t^3 - 2t + 1}{t^4 + 3}\right)^{12} \frac{-t^6 + 6t^4 - 4t^3 + 9t^2 - 6}{(t^4 + 3)^2}
\end{aligned}$$

Derivative

Chain Rule

EXAMPLE 4 If $y = \sin 2x$, find $\dfrac{dy}{dx}$.

SOLUTION The last step in calculating this expression would be to take the sine of the quantity $2x$. Thus we use the Chain Rule on the function $y = \sin u$ where $u = 2x$.

$$\frac{dy}{dx} = (\cos 2x)\left(\frac{d}{dx}2x\right) = 2\cos 2x \qquad \blacksquare$$

EXAMPLE 5 Find $F'(y)$ where $F(y) = y \sin y^2$.

SOLUTION The last step in calculating this expression would be to multiply y and $\sin y^2$, so we begin by applying the Product Rule. The Chain Rule is needed when we differentiate $\sin y^2$.

$$\begin{aligned} F'(y) &= yD_y[\sin y^2] + (\sin y^2)D_y(y) \\ &= y(\cos y^2)D_y(y^2) + (\sin y^2)(1) \\ &= 2y^2 \cos y^2 + \sin y^2 \qquad \blacksquare \end{aligned}$$

Derivative

Logarithmic Function

$$\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx}(\ln u) = \frac{1}{u}\frac{du}{dx}$$

$$\frac{d}{dx}[\ln g(x)] = \frac{g'(x)}{g(x)}$$

$$\frac{d}{dx}\ln|x| = \frac{1}{x}$$

V EXAMPLE 1 Differentiate $y = \ln(x^3 + 1)$.

SOLUTION To use the Chain Rule, we let $u = x^3 + 1$. Then $y = \ln u$, so

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = \frac{1}{u}\frac{du}{dx}$$

$$= \frac{1}{x^3 + 1}(3x^2) = \frac{3x^2}{x^3 + 1}$$

EXAMPLE 2 Find $\frac{d}{dx}\ln(\sin x)$.

SOLUTION

$$\frac{d}{dx}\ln(\sin x) = \frac{1}{\sin x}\frac{d}{dx}(\sin x) = \frac{1}{\sin x}\cos x = \cot x$$

Derivative

Logarithmic Function

EXAMPLE 3 Differentiate $f(x) = \sqrt{\ln x}$.

SOLUTION This time the logarithm is the inner function, so the Chain Rule gives

$$f'(x) = \tfrac{1}{2}(\ln x)^{-1/2} \frac{d}{dx}(\ln x) = \frac{1}{2\sqrt{\ln x}} \cdot \frac{1}{x} = \frac{1}{2x\sqrt{\ln x}}$$

EXAMPLE 4 Differentiate $f(x) = \log_{10}(2 + \sin x)$.

SOLUTION Using Formula 1 with $a = 10$, we have

$$\begin{aligned} f'(x) &= \frac{d}{dx}\log_{10}(2 + \sin x) \\ &= \frac{1}{(2 + \sin x)\ln 10}\frac{d}{dx}(2 + \sin x) \\ &= \frac{\cos x}{(2 + \sin x)\ln 10} \end{aligned}$$

EXAMPLE 5 Find $\frac{d}{dx} \ln \frac{x+1}{\sqrt{x-2}}$.

SOLUTION 1

$$\frac{d}{dx} \ln \frac{x+1}{\sqrt{x-2}} = \frac{1}{\frac{x+1}{\sqrt{x-2}}} \frac{d}{dx} \frac{x+1}{\sqrt{x-2}}$$

$$= \frac{\sqrt{x-2}}{x+1} \frac{\sqrt{x-2} \cdot 1 - (x+1)\left(\frac{1}{2}\right)(x-2)^{-1/2}}{x-2}$$

$$= \frac{x - 2 - \frac{1}{2}(x+1)}{(x+1)(x-2)}$$

$$= \frac{x-5}{2(x+1)(x-2)}$$

EXAMPLE 6 Find $f'(x)$ if $f(x) = \ln|x|$.

SOLUTION Since

$$f(x) = \begin{cases} \ln x & \text{if } x > 0 \\ \ln(-x) & \text{if } x < 0 \end{cases}$$

it follows that

$$f'(x) = \begin{cases} \dfrac{1}{x} & \text{if } x > 0 \\ \dfrac{1}{-x}(-1) = \dfrac{1}{x} & \text{if } x < 0 \end{cases}$$

Thus $f'(x) = 1/x$ for all $x \neq 0$.

Derivative

Exponential Function

Derivative of the Natural Exponential Function

$$\frac{d}{dx}(e^x) = e^x$$

V EXAMPLE 8 If $f(x) = e^x - x$, find f' and f''. Compare the graphs of f and f'.

SOLUTION Using the Difference Rule, we have

$$f'(x) = \frac{d}{dx}(e^x - x) = \frac{d}{dx}(e^x) - \frac{d}{dx}(x) = e^x - 1$$

$$f''(x) = \frac{d}{dx}(e^x - 1) = \frac{d}{dx}(e^x) - \frac{d}{dx}(1) = e^x$$

Derivative

Implicit Method

EXAMPLE 1 Find dy/dx if $4x^2y - 3y = x^3 - 1$.

SOLUTION

Method 1 We can solve the given equation explicitly for y as follows:

$$y(4x^2 - 3) = x^3 - 1$$

$$y = \frac{x^3 - 1}{4x^2 - 3}$$

Thus,

$$\frac{dy}{dx} = \frac{(4x^2 - 3)(3x^2) - (x^3 - 1)(8x)}{(4x^2 - 3)^2} = \frac{4x^4 - 9x^2 + 8x}{(4x^2 - 3)^2}$$

Method 2 Implicit Differentiation We equate the derivatives of the two sides.

$$\frac{d}{dx}(4x^2y - 3y) = \frac{d}{dx}(x^3 - 1)$$

We obtain, after using the Product Rule on the first term,

$$4x^2 \cdot \frac{dy}{dx} + y \cdot 8x - 3\frac{dy}{dx} = 3x^2$$

$$\frac{dy}{dx}(4x^2 - 3) = 3x^2 - 8xy$$

$$\frac{dy}{dx} = \frac{3x^2 - 8xy}{4x^2 - 3}$$

$$\frac{dy}{dx} = \frac{3x^2 - 8xy}{4x^2 - 3} = \frac{3x^2 - 8x\dfrac{x^3 - 1}{4x^2 - 3}}{4x^2 - 3}$$

$$= \frac{12x^4 - 9x^2 - 8x^4 + 8x}{(4x^2 - 3)^2} = \frac{4x^4 - 9x^2 + 8x}{(4x^2 - 3)^2}$$

Derivative

Implicit Method

EXAMPLE 2 Find dy/dx if $x^2 + 5y^3 = x + 9$.

SOLUTION

$$\frac{d}{dx}(x^2 + 5y^3) = \frac{d}{dx}(x + 9)$$

$$2x + 15y^2\frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1 - 2x}{15y^2}$$

EXAMPLE 3 Find the equation of the tangent line to the curve

$$y^3 - xy^2 + \cos xy = 2$$

at the point $(0, 1)$.

SOLUTION For simplicity, let us use the notation y' for dy/dx. When we differentiate both sides and equate the results, we obtain

$$3y^2y' - x(2yy') - y^2 - (\sin xy)(xy' + y) = 0$$

$$y'(3y^2 - 2xy - x\sin xy) = y^2 + y\sin xy$$

$$y' = \frac{y^2 + y\sin xy}{3y^2 - 2xy - x\sin xy}$$

At $(0, 1)$, $y' = \frac{1}{3}$. Thus, the equation of the tangent line at $(0, 1)$ is

$$y - 1 = \tfrac{1}{3}(x - 0)$$

or

$$y = \tfrac{1}{3}x + 1$$

Derivative

Higher Order

Derivative	f' Notation	y' Notation	D Notation	Leibniz Notation
First	$f'(x)$	y'	$D_x y$	$\dfrac{dy}{dx}$
Second	$f''(x)$	y''	$D_x^2 y$	$\dfrac{d^2y}{dx^2}$
Third	$f'''(x)$	y'''	$D_x^3 y$	$\dfrac{d^3y}{dx^3}$
Fourth	$f^{(4)}(x)$	$y^{(4)}$	$D_x^4 y$	$\dfrac{d^4y}{dx^4}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
nth	$f^{(n)}(x)$	$y^{(n)}$	$D_x^n y$	$\dfrac{d^ny}{dx^n}$

Notations for Derivatives of $y = f(x)$

EXAMPLE 1 If $y = \sin 2x$, find d^3y/dx^3, d^4y/dx^4, and $d^{12}y/dx^{12}$.

SOLUTION

$$\frac{dy}{dx} = 2\cos 2x$$

$$\frac{d^2y}{dx^2} = -2^2 \sin 2x$$

$$\frac{d^3y}{dx^3} = -2^3 \cos 2x$$

$$\frac{d^4y}{dx^4} = 2^4 \sin 2x$$

$$\frac{d^5y}{dx^5} = 2^5 \cos 2x$$

$$\vdots$$

$$\frac{d^{12}y}{dx^{12}} = 2^{12} \sin 2x$$

Derivative

Time to participate

use $f'(x) = \lim_{h \to 0} [f(x+h) - f(x)]/h$ *to find*

$$F(x) = \frac{x-1}{x+1}$$

$$H(x) = \sqrt{x^2+4}$$

1

2

use $f'(x) = \lim_{t \to x} [f(t) - f(x)]/[t - x]$ *to find*

$$f(x) = \frac{x}{x-5}$$

$$f(x) = \frac{x+3}{x}$$

3

4

Derivative

Time to participate

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find $D_x y$

1. $$y = \frac{1}{2x} + 2x$$

2. $$y = 3x(x^3 - 1)$$

find $D_x y$

3. $$y = \frac{\sin x + \cos x}{\tan x}$$

4. $$y = \frac{x \cos x + \sin x}{x^2 + 1}$$

Derivative

Time to participate

find $D_x y$

1. $y = (3x - 2)^2(3 - x^2)^2$

2. $y = \cos^3\left(\frac{x^2}{1 - x}\right)$

find the equation of the tangent line

3. $y + \cos(xy^2) + 3x^2 = 4; (1, 0)$

4. $\sqrt{y} + xy^2 = 5; (4, 1)$

Derivative

Time to participate

find $D_x y$ by implicit differentiation.

1. $$x\sqrt{y+1} = xy + 1$$

2. $$xy + \sin(xy) = 1$$

find d^3y/dx^3.

3. $$y = \frac{3x}{1-x}$$

4. $$y = \sin(x^3)$$

SEE YOU NEXT WEEK !